

x is an interior point of X , that is, there exists $\varepsilon > 0$ such that

$$\|z - x\| \leq \varepsilon \implies z \in X.$$

It means that the (convex) ball of radius ε and centered at x is completely in X .

Consider an arbitrary direction $d \neq 0$, and the point

$$z = x + \varepsilon \frac{d}{\|d\|}.$$

The point z is at the intersection of the direction d and the border of the ball. As

$$\|z - x\| = \left\| \varepsilon \frac{d}{\|d\|} \right\| = \varepsilon,$$

the definition of the interior point guarantees that $z \in X$. By convexity of the ball, all points $x + \alpha d$ with

$$0 \leq \alpha \leq \frac{\varepsilon}{\|d\|}$$

also belong to the ball, and therefore, to X , proving that the direction d is feasible.

